

Supporting Teamwork in Mixed-Ability Virtual Collaboration

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Abstract

The increase in virtual collaboration has significantly reshaped how teams work together, offering greater flexibility for people with and without disabilities. While prior research has documented accessibility barriers for people with disabilities and strategies to navigate virtual collaboration, there remains a gap in understanding how these accessibility practices shape overall teamwork in mixed-ability teams. To address this gap, my doctoral research aims to design a virtual collaboration tool that fosters more inclusive teamwork in mixed-ability settings. To inform this design, I take a two-fold approach. First, I conducted a study that examined how accessibility practices influence holistic teamwork in mixed-ability virtual teams. Second, based on my findings, I plan to design and evaluate a virtual collaboration tool that supports more inclusive teamwork through AI-driven features. This work advances the design of virtual collaboration technologies that facilitate inclusive teamwork.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Accessibility**; • **Social and professional topics** → **People with disabilities**.

Keywords

virtual collaboration, teamwork, accessibility, mixed-ability

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1 Introduction

The global shift to remote work prompted by the COVID-19 pandemic has highlighted the necessity and benefits of virtual collaboration [5]. For people with disabilities, this shift potentially reduces physical barriers to participation and leverages tools equipped with accessibility features such as closed captioning, further supporting collaborative engagement [17, 19].

As a result, research in Human-Computer Interaction and Computer-Supported Cooperative Work increasingly examines how technology can facilitate remote collaboration. Accessibility research has examined virtual collaboration tools specifically within *mixed-ability teams*, comprising both disabled and non-disabled members. However, most studies primarily emphasize individual-level accessibility barriers and practices to support disabled participants' needs. For instance, Mack et al. [16] highlighted virtual meeting accommodations like speaking slowly for interpreters and verbalizing one's identity before speaking to support blind and low vision members. Moreover, research by Alharbi et al. [1] and Mack et al. [16] illustrated how power imbalances could disrupt consistent accommodation practices. Despite these valuable insights, there remain significant gaps in understanding how these accessibility practices shape overall teamwork in mixed-ability teams.

Teamwork—the interactions and relationships influencing team performance and cohesion [14]—is crucial for effective collaboration. Yet, accessibility remains largely unexplored in traditional teamwork frameworks [3, 6, 7, 15, 20]. Previous studies have explored teamwork dynamics across multicultural teams [4], teams of varied socioeconomic backgrounds [10], and linguistically diverse teams [12]. For example, Cagiltay et al. [4] found that communication style differences significantly impacted multicultural virtual team collaboration. However, there is a lack of an understanding of teamwork specifically in mixed-ability contexts.

Recent advancements in virtual collaboration tools reflect opportunities to create systems that actively shape how teams work together. In particular, the emergence of AI-driven features in virtual meetings, such as smart meeting summaries and automated task management [2], illustrates how these tools are beginning to take a more active role in guiding collaboration. However, it remains unknown how such virtual collaboration tools should be designed in the context of supporting inclusive teamwork.

To address this gap, my research aims to answer the following question: **What design of a virtual collaboration tool can support inclusive teamwork?**

To answer this question, my research agenda is two-fold:

- (1) Understanding how accessibility practices shape overall teamwork in mixed-ability teams (completed work), and
- (2) Designing and evaluating a virtual collaboration tool to support inclusive teamwork (ongoing and future work).

By centering accessibility within the broader context of teamwork, this research advances our understanding of how mixed-ability teams collaborate in virtual environments and how technology can be designed to support more inclusive practices. The insights from this study contribute to our knowledge of remote collaboration and the design of virtual tools that actively foster inclusive teamwork.

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2 Understanding Teamwork in Mixed-Ability Teams that Collaborate Virtually

To examine how accessibility practices shape overall teamwork in mixed-ability teams that collaborate virtually, I drew on foundational teamwork research. A comprehensive review by Kozlowski et al. [13] of decades of research in this field identifies consistently emerging concepts. From their work, I focus on three dimensions: productivity, participation, and camaraderie. I used these dimensions as a lens to understand teamwork in mixed-ability teams. I conducted 60-minute semi-structured interviews with 18 participants (12 with various types of disabilities and six without disabilities) across a range of professional and academic teams. I probed about participants' teamwork experiences within their mixed-ability teams.

Through these interviews, I found that accessibility practices were not merely an afterthought but a lever for effective teamwork. For example, ongoing efforts to understand one another's disabilities and access needs contributed to both individual and collective productivity while also deepening interpersonal rapport within teams. Non-disabled participants, in particular, described allyship as an ongoing process of negotiation. Specifically, they discussed how they learned how to offer support without overstepping and navigated interpersonal boundaries in order to contribute meaningfully to an inclusive team environment.

However, participants also described ongoing tensions between empathy and accountability when supporting team members' access needs. While accommodations such as flexible deadlines, reminders, and adjusted workloads were intended to accommodate the needs of team members, these practices sometimes resulted in delays and uneven participation. Despite their commitment to accessibility, participants found it difficult to maintain team morale when accommodations undermined a shared sense of responsibility.

From these findings, I contribute the following: First, I demonstrate how accessibility practices reinforce strong teamwork. I detail how accessibility practices are integral to existing facets of teamwork. Second, I identify design opportunities for virtual collaboration tools that can better support inclusive teamwork. For instance, a feature that facilitates regular team discussions about accessibility practices could help formalize accommodations, destigmatize support, and reinforce camaraderie and productivity.

This work emphasizes the need to reimagine accessibility practices not just as a means of supporting people with disabilities but also as a pathway to fostering strong teamwork.

3 Designing and Evaluating a Virtual Collaboration Tool to Support Inclusive Teamwork

Building on my understanding of mixed-ability teamwork, the next phase of my research focuses on designing and evaluating a virtual collaboration tool for virtual meetings to support inclusive teamwork. My previous findings emphasized that accessibility practices are integral to effective teamwork. Thus, I aim to create a virtual collaboration tool that addresses these needs at the team level.

This tool features an AI agent that supports virtual meetings, the primary mode of collaboration among participants in my previous

study. Research has shown that AI agents are well-suited for assisting with collaborative tasks within teams [8, 9, 21], as they can analyze real-time interactions and adapt to evolving team needs [18]. Below, I describe my ongoing and future work.

3.1 Designing and Evaluating Collaboration Tool with Participants with Mixed-Ability Experience (Ongoing Work)

I am currently exploring how a virtual collaboration tool should be designed to address and support the needs and challenges uncovered in my earlier study. I recruited participants with and without disabilities who had prior experience working in mixed-ability teams. To explore potential designs, I developed a set of design probes that illustrate possible features of the AI agent, such as issuing real-time reminders about accommodations, mediating conflicting access needs, or prompting post-meeting reflections. These probes span different phases of collaboration, specifically before, during, and after meetings, and vary in the agent's embodiment, visibility, and level of initiative.

By presenting these probes to participants with experience in mixed-ability teams, I aim to identify key design preferences, priorities, and trade-offs. I anticipate that this study will surface preferences related to initiative (e.g., proactive vs. reactive behavior), visibility (e.g., private notifications vs. public announcements), and control (e.g., ability to configure or opt into features). It may also surface tensions between supporting team inclusivity and avoiding disruptions to workflow. In addition, participants may offer insight into how they perceive the role of the agent within the team, whether as a tool, assistant, or team member.

Following this design exploration, I will conduct Wizard-of-Oz evaluation studies to simulate AI-driven features during virtual meetings. Participants will complete two collaborative tasks—one that is supported by AI features and one that is not. After each task, participants will respond to Likert scale questions assessing their perceptions of the features, including their effectiveness and impact on teamwork. Additionally, qualitative feedback will be collected through semi-structured interviews to understand participants' experiences. This evaluation phase will provide insights into how AI features affect team collaboration, including perceptions of inclusivity. Based on these findings, I expect to outline design recommendations for AI-driven virtual collaboration tools that support inclusive practices in mixed-ability teams.

3.2 Evaluating Collaboration Tool with Participants without Prior Mixed-Ability Experience (Future Work)

In my future work, I plan to extend my evaluation to disabled and non-disabled participants *without* prior experience in mixed-ability collaboration. This future study will examine whether our virtual collaboration tool can support team members with little or no prior knowledge about accessibility to adopt inclusive collaboration practices. Evaluating the tool in this broader context will provide essential insights into its generalizability, demonstrating whether and how our virtual collaboration tool design can benefit not only team members already familiar with accessibility practices but also

those without prior experience. This evaluation will thus expand our understanding of the tool's applicability across diverse team compositions and organizational settings.

Moreover, this evaluation will shed light on the tool's potential to serve as an educational resource that raises team members' awareness of accessibility during collaboration. Prior research has shown that AI agents can enhance user engagement, comprehension, and learning outcomes [11]. Building on this, I aim to understand whether the tool can promote inclusive practices and foster accessibility awareness among team members without prior mixed-ability collaboration experience. Based on these findings, I expect to outline design recommendations for virtual collaboration tools that are effective for users without prior experience in mixed-ability teamwork. Ultimately, this research will inform the design of AI-driven collaboration tools and advance our understanding of how such tools can foster inclusive teamwork.

4 Contributions & Outcomes

The contributions of my research include: (1) an empirical understanding of how accessibility practices shape holistic teamwork in mixed-ability teams collaborating in virtual settings, and (2) the design and evaluation of a virtual collaboration tool with AI-driven features that foster inclusive teamwork. My work aligns with research in Human-Computer Interaction, Human-AI Interaction, and Computer-Supported Cooperative Work by adopting user-centered design principles, centering the lived experiences of both disabled and non-disabled team members, and designing AI systems that actively promote inclusive practices. More broadly, it contributes to ongoing conversations about virtual teamwork, the role of AI in collaboration, and the future of inclusive group work. The outcomes of this research offer actionable insights for HCI researchers, designers, and developers by providing design recommendations that position accessibility as a fundamental aspect of effective collaboration. Overall, this work will deepen our understanding of how to design inclusive collaboration technologies.

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